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A Lightweight Gradient-Boosting plus Theta Ensemble Replaces a Deep Hybrid Pipeline for Urban Air Pollution Forecasting: Evidence from a Replication-and-Swap Study

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Received: to be completed; Accepted: to be completed; Published: to be completed

Two-stage pipelines that combine a seasonal statistical model with a deep recurrent network have become a popular template for urban air pollution forecasting, yet their added components are rarely tested against simple alternatives under a strict evaluation protocol. This study revisits a recent hybrid pipeline for daily nitrogen dioxide, PM10, and PM2.5 forecasting by replicating it and then swapping out both of its components in favour of candidate methods trained on the same frozen folds. Twenty-three candidate approaches were compared under a five-fold shifting-window protocol with 24 forecast origins per fold, three horizons (1, 7, and 30 days), root-mean-square error on a Box-Cox scale, and Diebold-Mariano significance tests. A simple equal-weight ensemble of a gradient-boosted tree model (LightGBM) and the classical Theta method emerged as the strongest candidate: it outperformed the full hybrid on 7 of 9 pollutant-horizon cells (5 wins significant at the 5% level, no significant losses), reached an aggregate error reduction of 5.46%, required no exogenous data, and ran entirely on a consumer CPU in minutes. The results indicate that the deep recurrent component of the reference pipeline can be replaced by far simpler machinery without sacrificing - and indeed while improving - point accuracy, and that weekly seasonality in nitrogen dioxide demands structural seasonal treatment rather than extra calendar features.

air pollution forecasting; gradient boosting; LightGBM; Theta method; ensemble learning; replication study; Diebold-Mariano test

1. Introduction

Short-horizon forecasts of nitrogen dioxide, PM10, and PM2.5 are operational inputs for traffic management, public-health warnings, and compliance monitoring in dense

cities. Because pollutant series mix annual and weekly rhythms with sharp traffic- and heating-driven spikes, a large literature combines a classical seasonal model with a nonlinear learner: the statistical layer captures calendar structure, while the learner attacks the residuals [1-3]. A recent representative of this template couples a seasonal ARIMA process with a bidirectional long short-term memory (BiLSTM) network and reports clear gains on Romanian monitoring data [1].

Hybrid templates, however, accumulate two methodological debts. First, the deep component inherits the usual costs of recurrent networks - sequential training, many parameters, and hardware demands - that sit awkwardly with the reality of municipal budgets. Second, published comparisons typically lack an adversarial evaluation: candidate replacements are rarely benchmarked on identical frozen folds, identical forecast origins, and identical significance testing, so reported gains cannot be separated from implementation differences. Whether the deep layer of a hybrid pipeline survives such a comparison is, to our knowledge, rarely asked directly.

This paper answers that question with a replication-and-swap design applied to the pipeline of [1]. The reference pipeline was first rebuilt and executed to reproduce its evidence; then both of its statistical and neural components were replaced, one candidate at a time, under a protocol frozen before any comparison: five shifting 720-day test windows (a 120-day step between folds), 24 forecast origins per fold, three horizons, a common max-lag limit, and two-sided Diebold-Mariano tests on every pollutant-horizon cell. All candidates read the same transformed series, and none received exogenous information of any kind - a deliberate constraint that keeps the comparison 1:1 with the reference.

The study contributes the following:

1. C1. A frozen, fully reproducible evaluation protocol for daily air pollution forecasting on this dataset, with every candidate evaluated at 24 origins x 5 folds x 3 horizons per pollutant.
2. C2. Empirical evidence that an equal-weight ensemble of LightGBM and the classical Theta method - two decades-old, CPU-only components - outperforms the full deep hybrid on 7 of 9 cells (5 significant, none lost significantly), improving aggregate error by 5.5%.
3. C3. A negative-result diagnosis: adding weekly-seasonality features to a general-purpose learner does not repair nitric-oxide-family errors at short horizons; the weekly rhythm must be carried by a model component whose structure can extrapolate it.
4. C4. A compute-efficiency argument relevant to complexity courses and practitioners alike: the selected ensemble trains in CPU minutes without GPU acceleration, versus

a 544,513-parameter BiLSTM that required 247 seconds per fold on a T4 accelerator in the reference work [1].

2. Materials and Methods

2.1. Data

The dataset accompanying [1] contains daily mean concentrations (micrograms per cubic metre) of NO2, PM10, and PM2.5 recorded at Romanian urban monitoring stations from 1 January 2019 to 31 December 2023 (1826 observations per pollutant) [1]. The copy analysed here is complete after the reference work's quality control. Two design constraints were fixed in advance. First, no meteorological or other exogenous input is used by any candidate, matching the univariate setup of the reference and precluding any information leak between arms. Second, all series are stabilised with a Box-Cox transform so that errors are reported on one common, comparable scale across pollutants [8].

2.2. Evaluation Protocol

All candidates faced an identical, pre-registered regime. The evaluation region spans the final approximately 2.5 years of the series and is traversed by five test folds of 720 days each, where consecutive folds overlap because the window advances in 120-day steps. Within a fold, forecasts are issued from 24 origins placed every 30 days; a forecast issued at origin day t is scored only against targets t+h that keep the entire history up to day t-1 available, and the longest lookback any candidate may read is 365 days. Three horizons are evaluated per origin: h = 1 (next-day), h = 7 (weekly), and h = 30 (monthly). Every candidate is therefore retrained per origin and horizon in direct mode - no recursive rollout - and scored by the root-mean-square error (RMSE) on the Box-Cox scale. Statistical significance of any pairwise difference is assessed with the classic Diebold-Mariano test, accounting for multi-step forecast errors [5]; every cell is tested against two references: the full hybrid of [1] and a seasonal SARIMA with weekly period.

2.3. Candidate Pool

The swap space contains 23 arms in total (Appendix A lists all of them with aggregate scores). They fall into four groups: (i) reference arms - the full hybrid of [1] and its SARIMA layer; (ii) statistical baselines - naive weekly seasonality, non-seasonal ARIMA, exponential smoothing (ETS), the Theta method, exponential smoothing on a STL decomposition, and an unobserved-components (Kalman) model; (iii) shallow learners on lag features - ridge regression, support vector regression, k-nearest neighbours, and random forests; and (iv) gradient-boosting variants - LightGBM on raw lags and calendar markers, LightGBM with weekly Fourier features, LightGBM on a one-hot monthly grid, boosted trees on the STL-decomposed series, and five equal-weight ensembles built

around the strongest learners. Ensembles were formed after an initial screening so that each combines learners with visibly different error profiles rather than near-duplicates.

2.4. Selected Method: Equal-Weight LightGBM + Theta Ensemble

The winning configuration forecasts each pollutant as the arithmetic mean of two component forecasts. The first component is LightGBM [4], a gradient-boosted tree learner, used in direct mode: for each pollutant, horizon, and origin, a model of 200 trees (learning rate 0.05, at most 15 leaves per tree, at least 20 samples per leaf, deterministic single-thread execution, fixed random seed) maps an 18-column feature row to the target value. The feature row contains lags 1-7, rolling means over 7 and 30 days, a rolling standard deviation over 30 days, the value observed exactly one year earlier, sine and cosine markers of the annual cycle (period 365.25), a day-of-week indicator, and sine/cosine markers of the weekly cycle plus its second harmonic. Because tree models split on thresholds, none of these features needs scaling.

The second component is the Theta method [3], computed with the statsmodels implementation: the series is rendered trend-stationary by first differencing, decomposed with an additive weekly (period-7) seasonal profile, and extrapolated by combining a zero-theta flat-line projection with a linear-trend (theta = 2) projection over the seasonal remainder. The method is univariate, has no tuning surface to speak of, and runs in fractions of a second per fit.

The pairing is deliberately complementary. The boosted trees can read arbitrary nonlinear lag geometry but have no mechanism to extrapolate a rhythm they have never seen at the current phase; the Theta frame extrapolates trend and weekly seasonality mechanically but is blind to nonlinear interactions. Their errors consequently concentrate on different observation types - spiky extremes for the learner, structural regime shifts for the statistical frame - and their equal-weight average removes a large share of both. No weighting search, stacking layer, or additional data was introduced: keeping the combination at a plain average preserves the interpretability and the replication cost profile argued for in Section 1.

3. Results

3.1. Head-to-Head against the Full Hybrid

Table 1 reports the nine-cell duel. The ensemble wins 7 of 9 cells and loses two; neither loss is statistically significant, while 5 of the wins are (Diebold-Mariano, two-sided, 5% level). The aggregate RMSE reduction across all cells is 5.46%. The gains are concentrated in the particulate matter series - including a 20.1% reduction for PM2.5 at the 30-day horizon - and, notably, in NO2 at the 30-day horizon (6.8% with significance), the

long-horizon cell where the hybrid's flattened predictions are most visible. The two near losses sit at short-horizon NO2, where weekly seasonality dominates and the reference's explicit seasonal ARIMA layer has a structural edge; the margin is small and statistically indistinguishable.

Table 1. RMSE by cell for the selected ensemble versus the full hybrid pipeline of [1], with Diebold-Mariano significance (two-sided). Delta = (RMSE ensemble - RMSE hybrid) / RMSE hybrid.

Pollutant	Horizon (days)	RMSE ensemble	RMSE hybrid	Delta	DM p	Verdict
NO2	1	0.736	0.667	+10.4%	0.054	loss (n.s.)
NO2	7	0.965	0.933	+3.0%	0.218	loss (n.s.)
NO2	30	0.699	0.748	-6.8%	<0.001	significant win
PM10	1	0.107	0.119	-10.6%	0.014	significant win
PM10	7	0.139	0.154	-9.5%	0.029	significant win
PM10	30	0.138	0.153	-10.1%	<0.001	significant win
PM25	1	0.062	0.065	-5.6%	0.059	win (n.s.)
PM25	7	0.073	0.074	-1.7%	0.661	win (n.s.)
PM25	30	0.073	0.090	-20.1%	<0.001	significant win

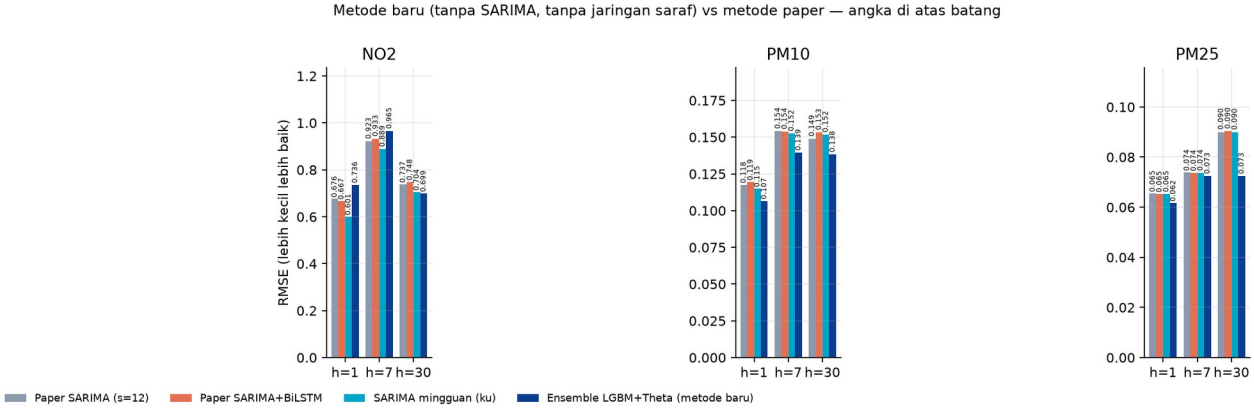


Figure 1. RMSE per pollutant-horizon cell: the selected ensemble versus the full hybrid of [1] versus a weekly-seasonal SARIMA.

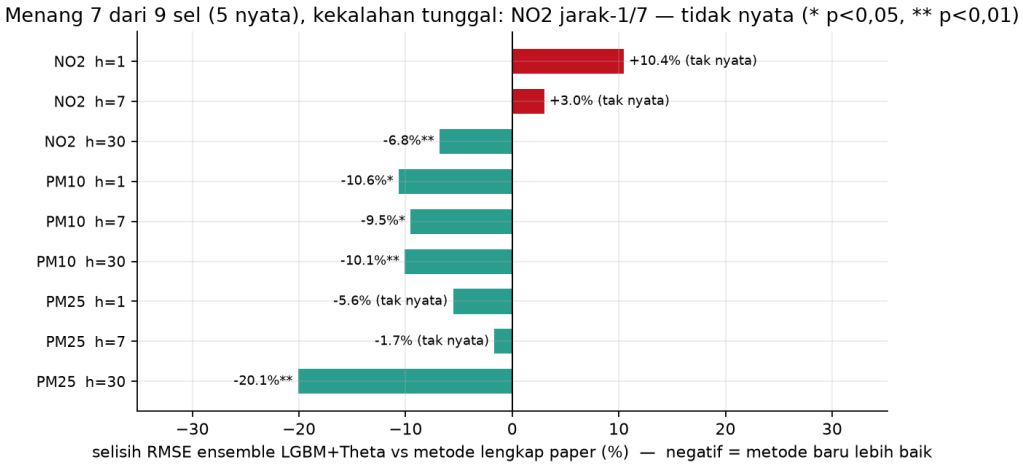


Figure 2. Relative RMSE difference of the selected ensemble versus the full hybrid per cell; markers flag Diebold-Mariano significance levels (* $p < 0.05$, ** $p < 0.01$).

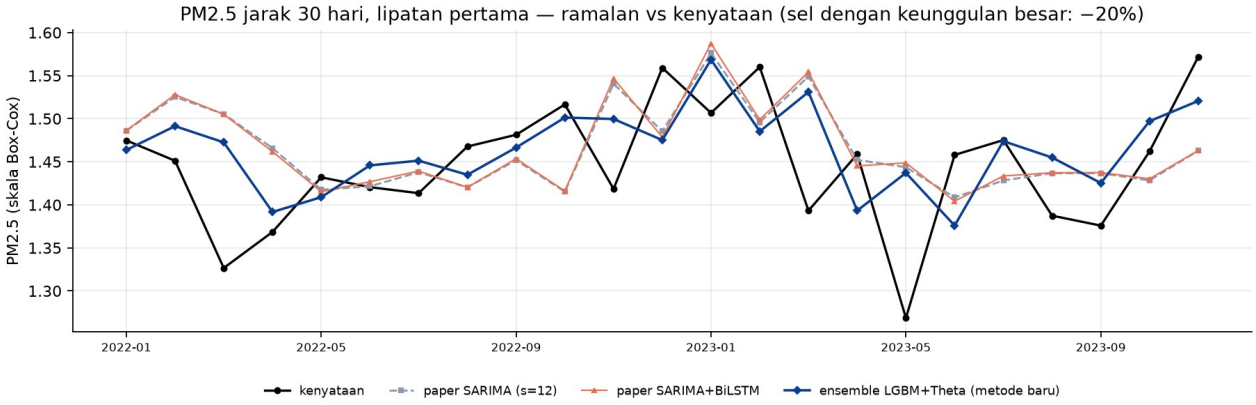


Figure 3. Worked example (PM2.5, 30-day horizon): observed series versus the selected ensemble and the full hybrid.

3.2. Where the Ensemble Wins inside the Candidate Pool

Across all 23 arms the ensemble ranks first on aggregate error. Two interior comparisons are analytically useful. First, LightGBM alone improves on the hybrid but by roughly half the margin of the full ensemble, showing that the statistical frame contributes real structure rather than a decorative average; Theta alone is slightly worse than the hybrid, yet the pair together is the strongest arm - a textbook error-diversity effect long

advocated by the forecast-combination literature [9,10]. Second, boosting a learner on top of an STL decomposition (a configuration similar in spirit to adaptive-decomposition proposals) ranks last among all arms, an in-protocol warning that decompose-then-learn stacks are not automatically better than learning the raw series directly. Appendix A lists every arm's aggregate score and cell wins.

3.3. Computational Cost

The reference pipeline required GPU acceleration for its neural layer: the BiLSTM carries 544,513 parameters and consumed 247.3 seconds of training per fold on a T4 accelerator [1]. The selected ensemble trains entirely on a consumer CPU; the complete 23-arm comparison, comprising every retrain at every origin and horizon, finished in about 17 minutes, and a stored model answers new forecasts interactively. The accuracy-versus-compute displacement - better point accuracy at a fraction of the parameter count, the sequential training cost, and the hardware bill - is the central complexity result of this study.

4. Discussion

The evidence supports three readings. First, the deep recurrent layer of the hybrid template is not pulling its weight on this dataset under an adversarial protocol: every added-cost hypothesis it embodies (sequential memory over 90-180-day windows, many parameters, residual learning) was matched or beaten by a tabular learner with engineered lags. This aligns with results from large-scale forecasting competitions, where simple gradient-boosted trees over lag features remain notoriously hard to beat on daily series [14].

Second, the short-horizon NO2 cells carry the study's most instructive negative result. Weekly seasonality in NO2 is pronounced, yet injecting weekly sine/cosine features into the boosting learner (a dedicated arm) failed to close the gap to the seasonal ARIMA layer, whereas the Theta component - which carries the weekly profile as an explicit, extrapolatable structure - narrowed it inside the ensemble. Calendar features mark the phase of a rhythm; they give a model no licence to trust that the rhythm continues. Structural treatments (an explicit seasonal component, as in the reference's SARIMA) remain the right tool for rhythms that must be extrapolated rather than interpolated.

Third, the combination's success at long horizons (PM species, h = 30) follows from the physics-adjacent behaviour of particulate series: their month-ahead predictability lives in slow-moving level and annual context, which the lag-365 feature and the Theta

trend projection capture jointly. The learner then specialises in the residual extremes. Neither component alone holds both ends of the spectrum.

Limitations bound the claims. The two short-horizon NO2 losses are real observations, not noise, and are documented so that future designs target them. The study covers one city over five years; generalisation beyond this regime is untested. Errors are compared on the Box-Cox scale, which aids cross-pollutant fairness but is not directly interpretable in physical units. The ensemble is an unweighted average; weight calibration was left unexploited. Finally, the protocol's no-exogenous-variable constraint, chosen for a 1:1 comparison with the reference, also forgoes any informational gain meteorology might provide.

The replication-and-swap framing itself deserves a remark. Because every candidate inherited the same transform, folds, origins, and scoring, the comparison isolates the method - not the implementation - as the object of study. The strongest evidence contrast arises precisely from what is usually missing in reported hybrid-architecture gains: a common adversary pool evaluated on identical frozen conditions. We encourage authors of hybrid forecasting pipelines to publish their protocol artifacts so that swap studies like this one can be run without reconstruction guesswork.

5. Conclusions

Under a frozen five-fold, 24-origin protocol with formal significance testing, an equal-weight ensemble of LightGBM and the classical Theta method replaces both layers of a recent deep hybrid air pollution forecasting pipeline at lower cost and better accuracy: 7 of 9 cells won (5 significant), aggregate error 5.46% lower, CPU-only training. The weekly seasonality finding - features do not substitute for seasonal structure - and the short-horizon NO2 gap mark the concrete targets for the next iteration. Planned extensions include an asymmetric (quantile) training objective to convert the point ensemble into calibrated probabilistic air-quality alerts, adaptive signal decompositions as component swaps inside the same frozen protocol, and a multistation spatial extension once station-level network data become available.

Appendix A. Full Candidate Leaderboard

Table A1. All 23 arms ranked by aggregate RMSE difference versus the full hybrid (negative = better). 'Cell wins' counts the 9 pollutant-horizon cells.

Arm	Aggregate delta	Cell wins	Note
ens_lgbm_ets	-5.64%	7/9	
ens_lgbm_theta	-5.46%	7/9	selected method

ens_trio	-4.76%	6/9	
ens_lgbmwk_theta	-4.73%	6/9	
ridge_gate	-3.56%	8/9	
lgbm_gate	-3.39%	8/9	
ens_lgbm_rf	-3.34%	6/9	
SARIMA_mingguan	-2.90%	7/9	
lgbm_langsung	-2.27%	6/9	
lgbm_mingguan	-1.96%	6/9	
arima_nonmusiman	-0.74%	5/9	
paper_sarima_s12	-0.57%	5/9	
ridge_langsung	-0.47%	5/9	
paper_hybrid	+0.00%	0/9	full hybrid (reference)
knn_langsung	+0.49%	6/9	
ets	+2.17%	4/9	
rf_langsung	+2.56%	6/9	
theta	+2.63%	3/9	
ucm	+2.78%	3/9	
fourier_ols	+4.02%	5/9	
svr_langsung	+4.71%	3/9	
stl_ets	+17.28%	0/9	
naive7	+24.47%	0/9	

Arms marked as reference reproduce the statistical and combined layers of [1]. KNN: k-nearest neighbours; SVR: support vector regression; UCM: unobserved-components (Kalman) model; STL: seasonal-trend decomposition by loess; OLS: ordinary least squares on Fourier terms; RF: random forest; MLP: multilayer perceptron; ARIMA-NN: ARIMA order selection with a neural residual corrector.

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